

I. Data to Be Collected

a. Goal

Mission Objective: REQ.03.01 — The ground-penetrating radar sounder shall resolve ice shell thickness to within $\pm 10\%$ accuracy for ice layers between 1 km and 30 km depth, producing a subsurface map at a minimum spatial resolution of 100 m along the traverse track. This requirement is directly traceable to the overarching mission objective of autonomously identifying and reaching an optimal cryobot insertion site on Europa's surface.

Data Product Goal: The data product is a georeferenced subsurface ice-thickness map assembled from continuous ground-penetrating radar (GPR) sounder returns collected during surface traversal. It provides mission engineers and onboard autonomy with a spatially resolved picture of where the ice shell is thinnest along the planned route, enabling data-driven site selection for cryobot deployment without real-time Earth support. Scientists receive a secondary product — a dielectric permittivity profile — that constrains ice composition and the presence of brine layers that would affect cryobot thermal penetration rates.

Success Criterion: The map is considered mission-sufficient when it covers the full planned traverse track at no worse than 100 m along-track spatial resolution and reports ice thickness values with a demonstrated accuracy of $\pm 10\%$ against calibration targets across the 1–30 km depth range (REQ.03.01). The onboard site-selection algorithm must be able to ingest this map and autonomously designate a candidate insertion waypoint without a ground-in-the-loop decision.

b. Data Collected by the Robot

- **Category 1 — Radar Sounder Returns (Primary Science Data):**

- Two-way travel time of radar pulse to primary basal reflector (nanoseconds, ns)
- Received signal amplitude at basal reflector (dB relative to transmitted power)
- Estimated dielectric permittivity of bulk ice column (dimensionless, ϵ_r)
- Derived ice shell thickness from travel-time inversion (meters, m)

- **Category 2 — Rover Localization and Pose:**

- Rover position estimate from visual odometry and inter-agent ranging (meters, m; in local ENU frame)
- Position uncertainty ($1\text{-}\sigma$ ellipse semi-axes, meters)
- Rover heading (degrees, relative to local North)
- Accumulated odometry error flag (Boolean: within ± 0.5 m bound per REQ.02.01)

- **Category 3 — Surface Terrain Context:**

- Surface slope at measurement site (degrees)
- Terrain roughness index derived from stereo or lidar returns (dimensionless, 0–1 scale)
- Surface temperature at measurement site (Kelvin, K)
- Estimated surface albedo at measurement site (dimensionless, 0–1 scale)

- **Category 4 — System Health and Data Quality Flags:**

- Total ionizing dose accumulated by radar front-end electronics at time of measurement (krad)
- Radar sounder noise floor at time of measurement (dB)
- Inter-agent network link quality at time of measurement (packet loss rate, %)
- Data quality flag for each radar trace (enumerated: NOMINAL / CLUTTER / NOISE_FLOOR / INVALID)

- **Daily Summary:**

- Total traverse distance completed in sol (meters, m)
- Number of valid radar traces collected in sol (count)
- Minimum ice thickness value observed in sol (meters, m) and associated position
- Number of candidate insertion sites flagged by onboard site-selection algorithm (count)
- Mean consensus convergence time for terrain map updates in sol (minutes)

II. Types of Measurements on the Spreadsheet

The spreadsheet is organized into multiple sheets to keep the dataset clear and usable. Below is a breakdown of each sheet:

1. Sheet 1 — Radar_Traces:

One row per radar sounder measurement. Columns capture the along-track position (local X, Y coordinates in meters), two-way travel time (ns), received amplitude (dB), estimated permittivity ϵ_r , derived ice thickness (m), and per-trace data quality flag. This is the raw science record; every other sheet derives from it. Columns are locked after initial population to preserve data integrity. A permittivity lookup table embedded in the sheet converts travel time to thickness using the formula $d = \frac{c \cdot t}{2\sqrt{\epsilon_r}}$, expressed as an Excel formula referencing the measured travel-time and permittivity columns rather than hardcoded values.

2. Sheet 2 — Localization_Log:

One row per odometry update, timestamped and synchronized with radar trace indices for georeferencing. Columns include rover position (X, Y, m), heading (deg), 1- σ position uncertainty (m), and the Boolean localization error flag. This sheet provides the spatial backbone that maps each radar trace in Sheet 1 to a physical surface location. Cross-sheet INDEX/MATCH formulas in Sheet 1 pull the nearest position fix for each radar trace timestamp.

3. Sheet 3 — Terrain_and_Health:

One row per measurement epoch, aligned to the same trace index as Sheet 1. Captures surface slope (deg), terrain roughness index, surface temperature (K), accumulated TID (krad), radar noise floor (dB), and inter-agent link packet loss (%). This sheet contextualizes each radar measurement: high slope or high roughness values flag geometrically challenging measurement conditions that degrade thickness accuracy, while elevated TID or noise floor values inform science team confidence in the corresponding trace.

4. Sheet 4 — Daily Summary:

One row per sol of operations. Auto-populated via Excel formulas referencing Sheets 1–3: total traverse distance (SUM of incremental odometry steps), valid trace count (COUNTIF on

quality flag column), minimum observed ice thickness (MIN on derived thickness column for the sol), candidate site count (COUNTIF on a site-selection flag column), and mean consensus convergence time. This sheet is the primary operational report delivered to mission planning each sol and feeds directly into the onboard replan trigger logic.

IV. Rationale and Alignment with Mission Objective

Alignment with Objective: Every data category maps directly to a requirement in the mission requirements document. The radar sounder returns (Category 1) are the primary data product required by REQ.03.01: without two-way travel time and permittivity, no thickness inversion is possible and the cryobot insertion site cannot be selected. The localization log (Category 2) is required by REQ.02.01: each radar trace must be georeferenced to ± 0.5 m to produce a spatially accurate subsurface map. Without position, the sounder data is a series of unlocated depth readings rather than a map. Terrain context (Category 3) directly feeds the autonomous site-selection logic: surface slope and roughness constrain physically accessible insertion sites independent of ice thickness, and temperature informs thermal modeling for cryobot descent rate estimation. System health flags (Category 4) are required by REQ.01.02 and REQ.04.01: elevated TID accumulation can degrade radar front-end performance, and inter-agent link quality affects the timeliness of map consensus updates that drive traverse planning.

Why These Measurements Are Sufficient: The four categories together provide complete closure on REQ.03.01 and its supporting requirements. Two-way travel time plus permittivity produces thickness; localization pins it to a map coordinate; terrain context filters candidate sites by physical accessibility; health flags bound the confidence of every measurement. The daily summary sheet provides the aggregate statistics needed by the onboard planner to decide whether to continue traversing or commit to an insertion attempt. No additional measurements are required to make the site-selection decision autonomously.

Potential Gaps or Limitations: The primary limitation is that permittivity estimation from GPR returns alone carries uncertainty when the ice column contains brine inclusions or heterogeneous layering, as identified in RISK-05 of the requirements document. Brine lenses increase local permittivity and produce spurious reflections that could cause the thickness algorithm to underestimate depth to the basal interface. The data product mitigates this by flagging affected traces with the CLUTTER quality code, but it does not eliminate the ambiguity without a secondary instrument (e.g., a seismic sensor or a multi-frequency radar). A second limitation is that along-track spatial resolution of 100 m is defined by REQ.03.01 but cross-track coverage depends on the swarm's orbital relay scan width, which is not yet specified; the current dataset captures a 1D profile along the traverse, not a 2D areal map. Future mission iterations should consider expanding the swarm's orbital synthetic aperture to provide cross-track coverage before committing to a surface traverse direction.